

EELE 3332 – Electromagnetic II

Chapter 11

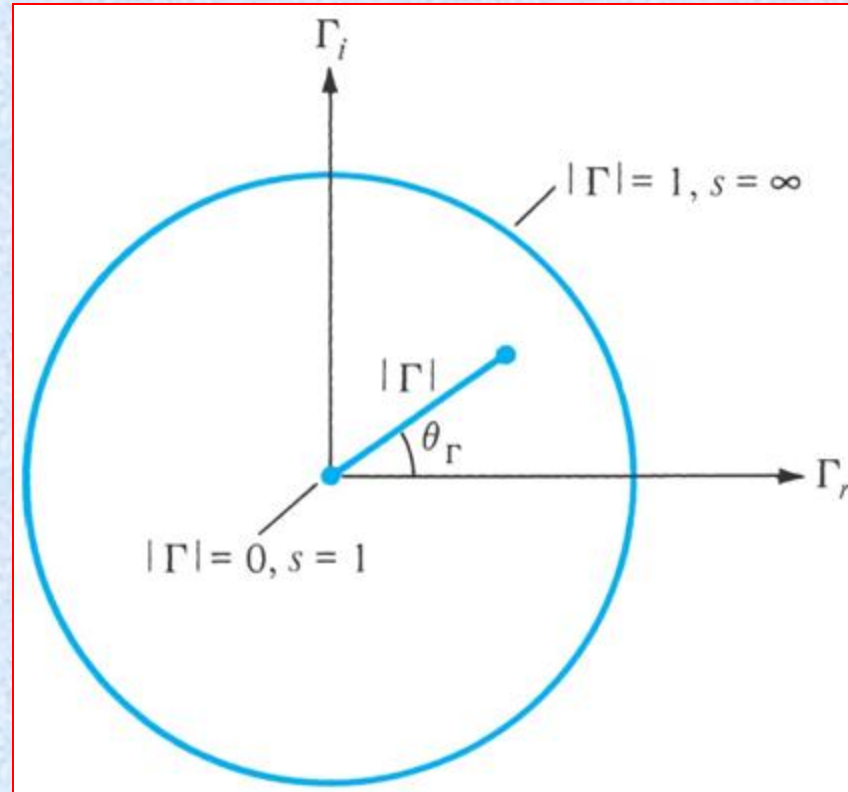
Transmission Lines

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11.5 The Smith Chart

- The Smith Chart is a graphical tool for analyzing transmission lines.
- The Smith Chart is made up of circles and arcs of circles.
- The circles are called “*constant resistance circles*”
- The arcs are “*constant reactance circles*”
- The combination of intersecting circles and arcs inside the chart allow us to locate the normalized impedance and then to find the impedance anywhere on the line.
- We will assume lossless transmission lines. ($Z_o = R_o$)

The Smith Chart



➤ The Smith Chart is constructed within a circle of unit radius ($|\Gamma| \leq 1$).

$$\Gamma = |\Gamma| \angle \theta_\Gamma = \Gamma_r + j\Gamma_i$$

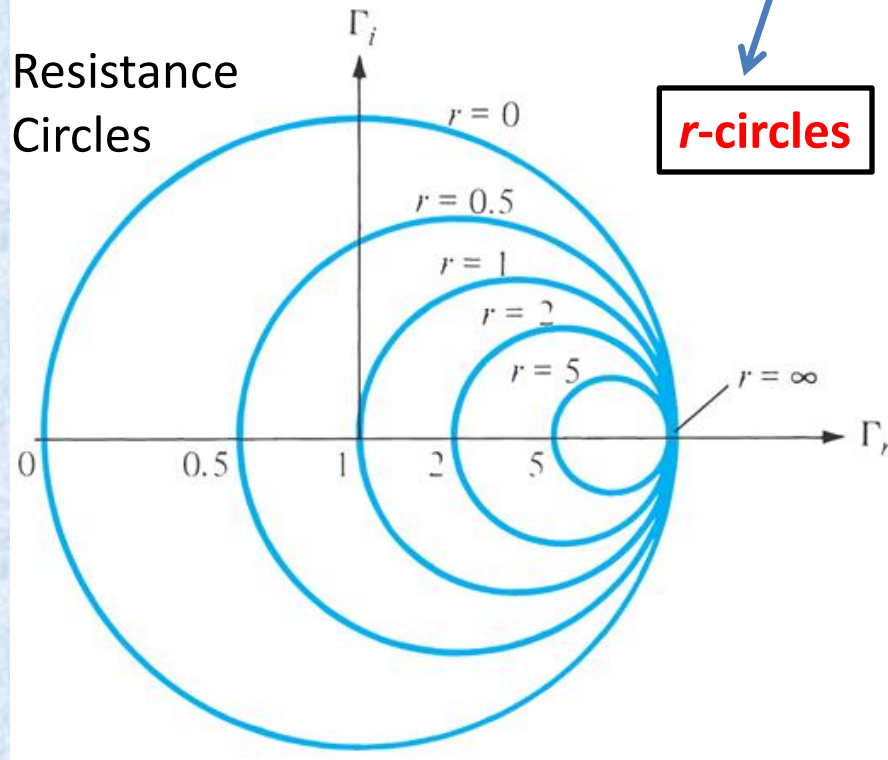
➤ Where Γ_r and Γ_i are the real and imaginary parts of Γ .

The Smith Chart

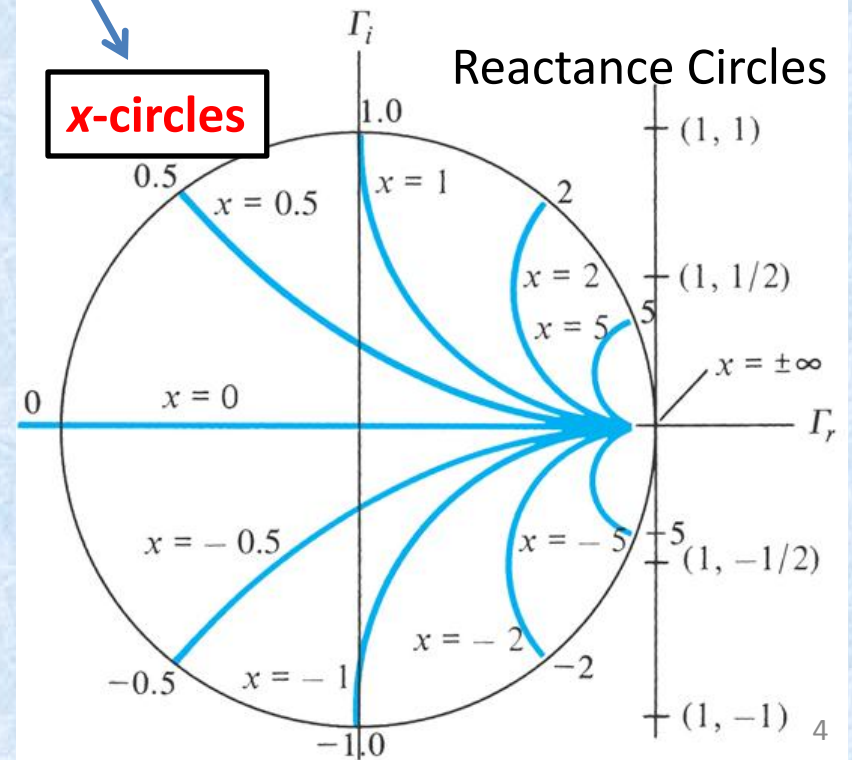
Smith Chart is normalized so that all impedances are normalized to the characteristic impedance Z_0 . (Hence, it can be used for any line)

For the load impedance Z_L , the normalized impedance z_L is:

$$z_L = \frac{Z_L}{Z_0} = r + jx$$

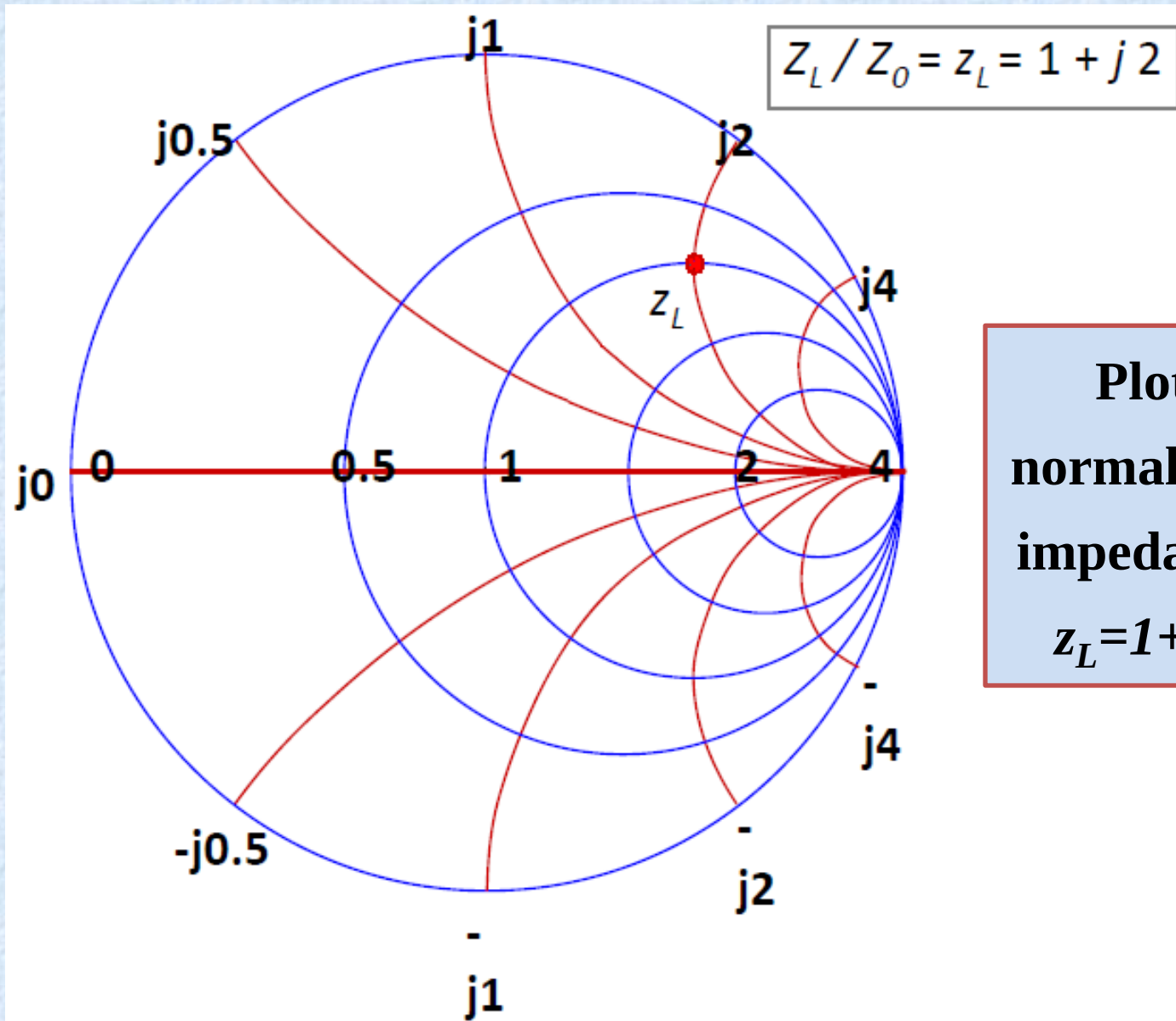


r-circles



x-circles

The Smith Chart



The Smith Chart

Example: Find Γ , given $Z = 25 + j100 \Omega$ with $Z_0 = 50 \Omega$

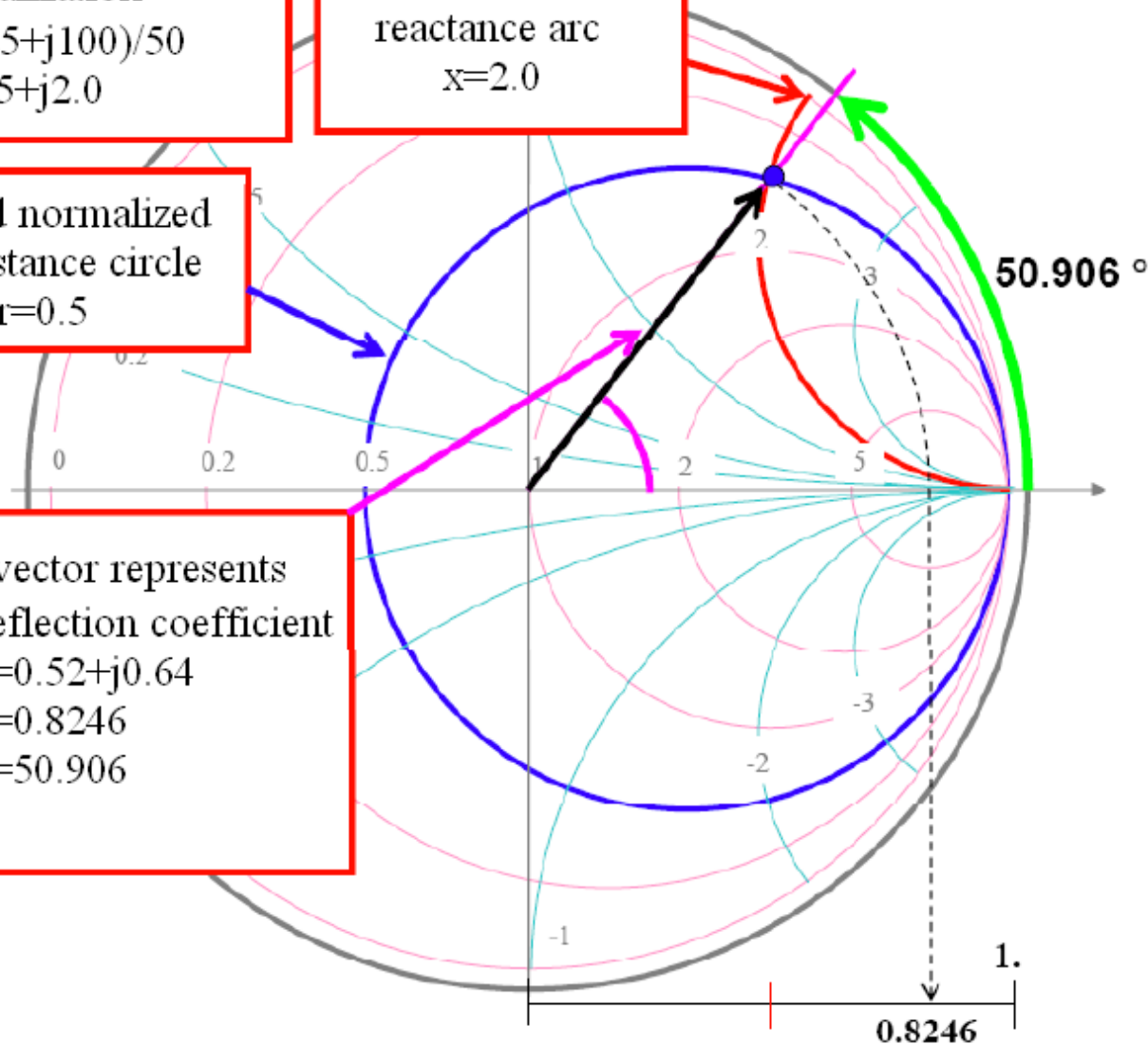
1. Normalization

$$Z_n = (25 + j100)/50 \\ = 0.5 + j2.0$$

3. Find normalized
reactance arc
 $x=2.0$

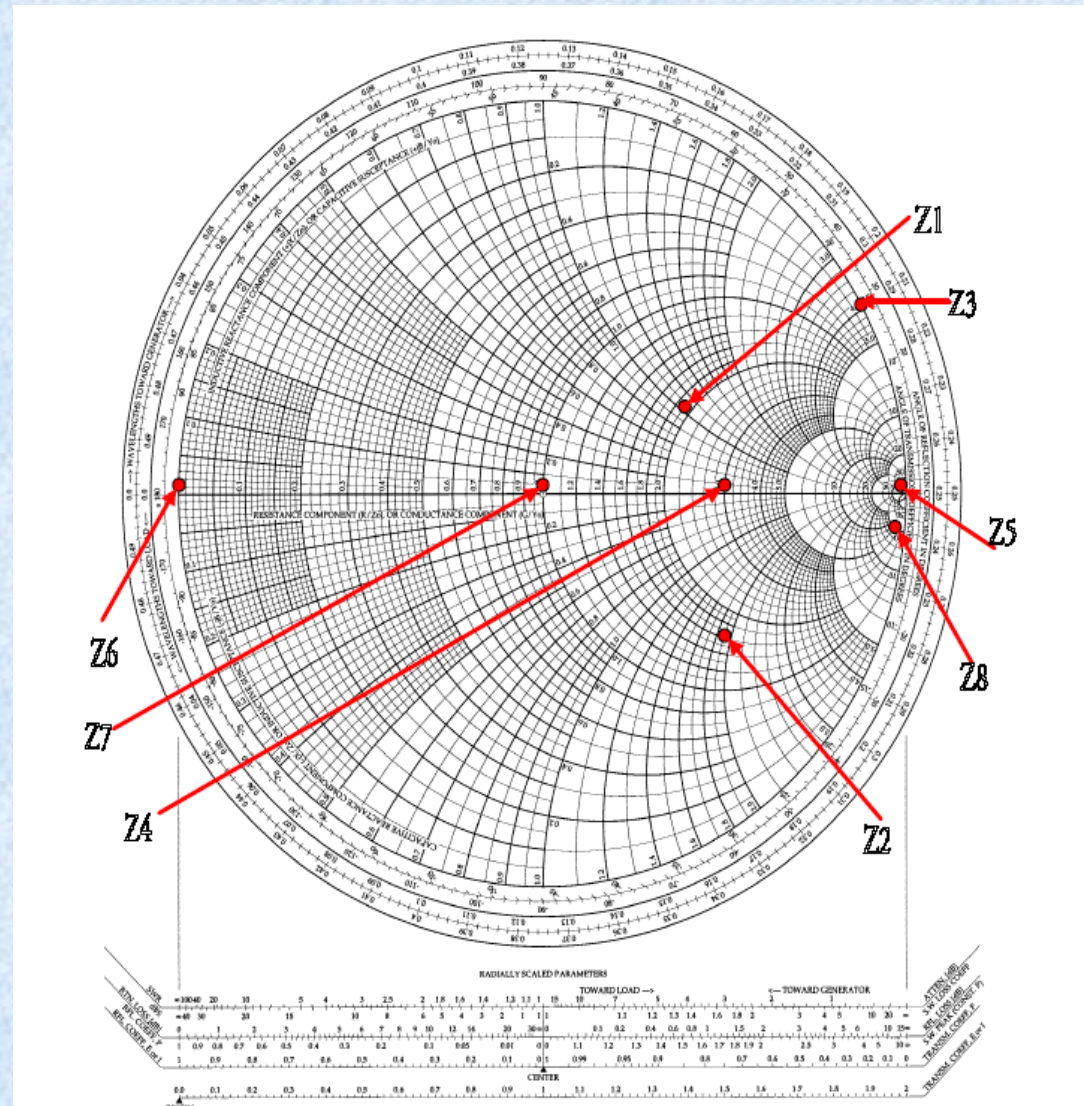
2. Find normalized
resistance circle
 $r=0.5$

4. This vector represents
the reflection coefficient
 $\Gamma = 0.52 + j0.64$
 $|\Gamma| = 0.8246$
 $\angle \Gamma = 50.906$

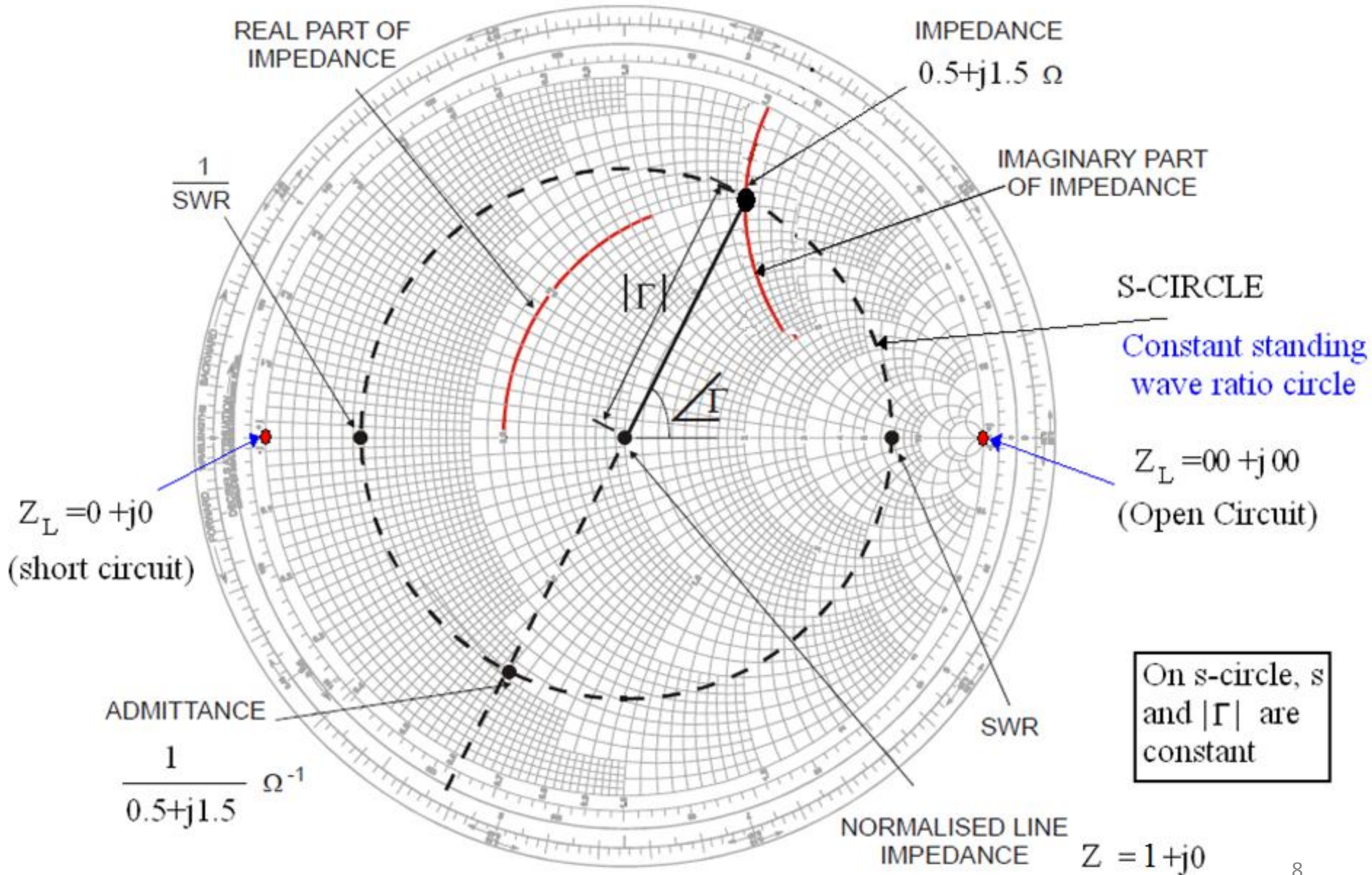


The Smith Chart

- Impedance:
 - $Z_1 = 100 + j50$
 - $Z_2 = 75 - j100$
 - $Z_3 = j200$
 - $Z_4 = 150$
 - $Z_5 = \text{infinity}$ (an open circuit)
 - $Z_6 = 0$ (a short circuit)
 - $Z_7 = 50$
 - $Z_8 = 184 - j900$
- Normalized impedance (line impedance 50 Ohms) :
 - $z_1 = 2 + j$
 - $z_2 = 1.5 - j2$
 - $z_3 = j4$
 - $z_4 = 3$
 - $z_5 = \text{infinity}$
 - $z_6 = 0$
 - $z_7 = 1$
 - $z_8 = 3.68 - j18$



The Smith Chart



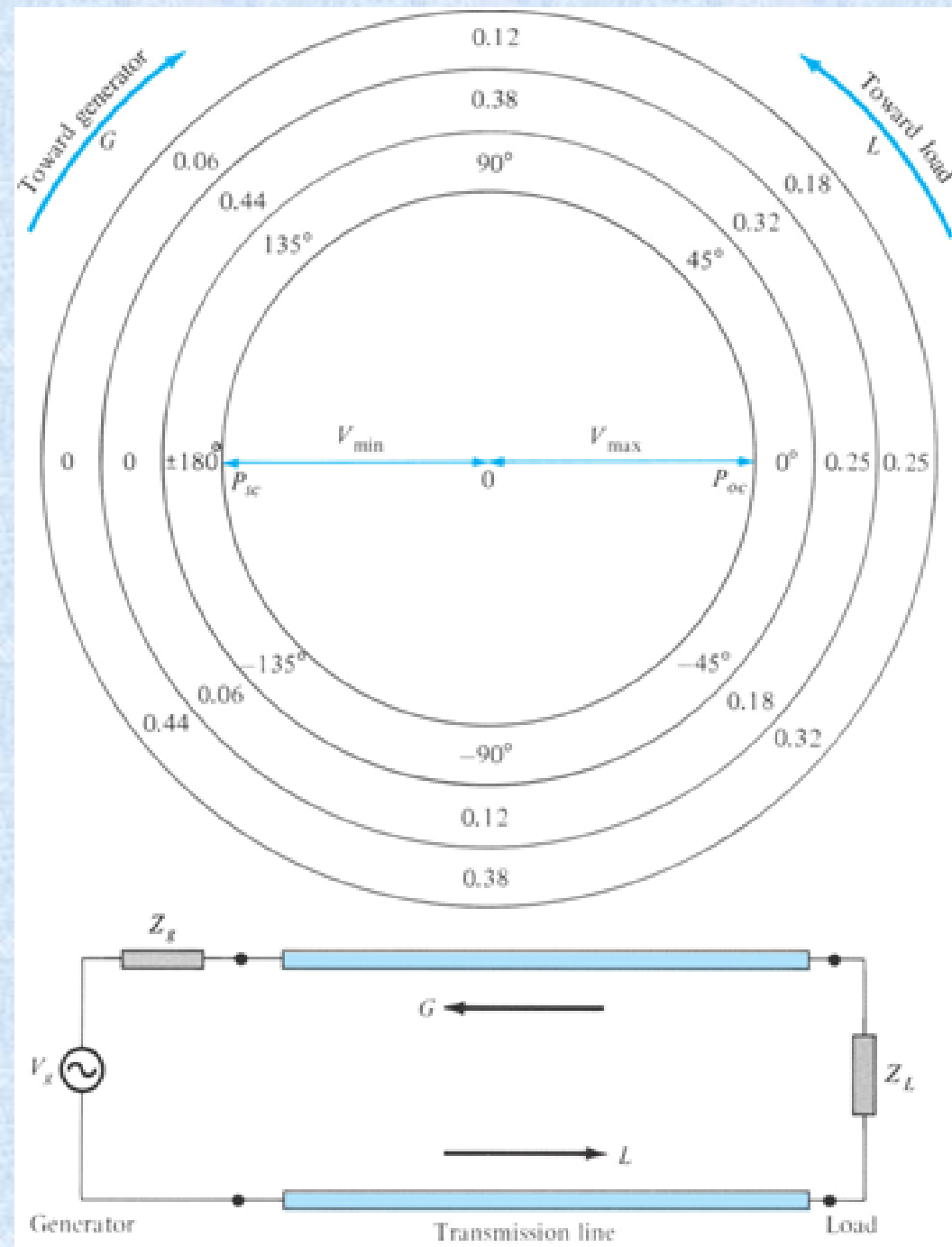
The Smith Chart

- A complete revolution (360°) around Smith Chart represents a distance of $(\lambda/2)$ on the line.

λ distance on the line corresponds to a 720° movement on the chart.

$$\lambda \rightarrow 720^\circ$$

- **Clockwise** movement represents moving toward the **Generator**.
- **Counter-clockwise** movement represents moving toward the **Load**.

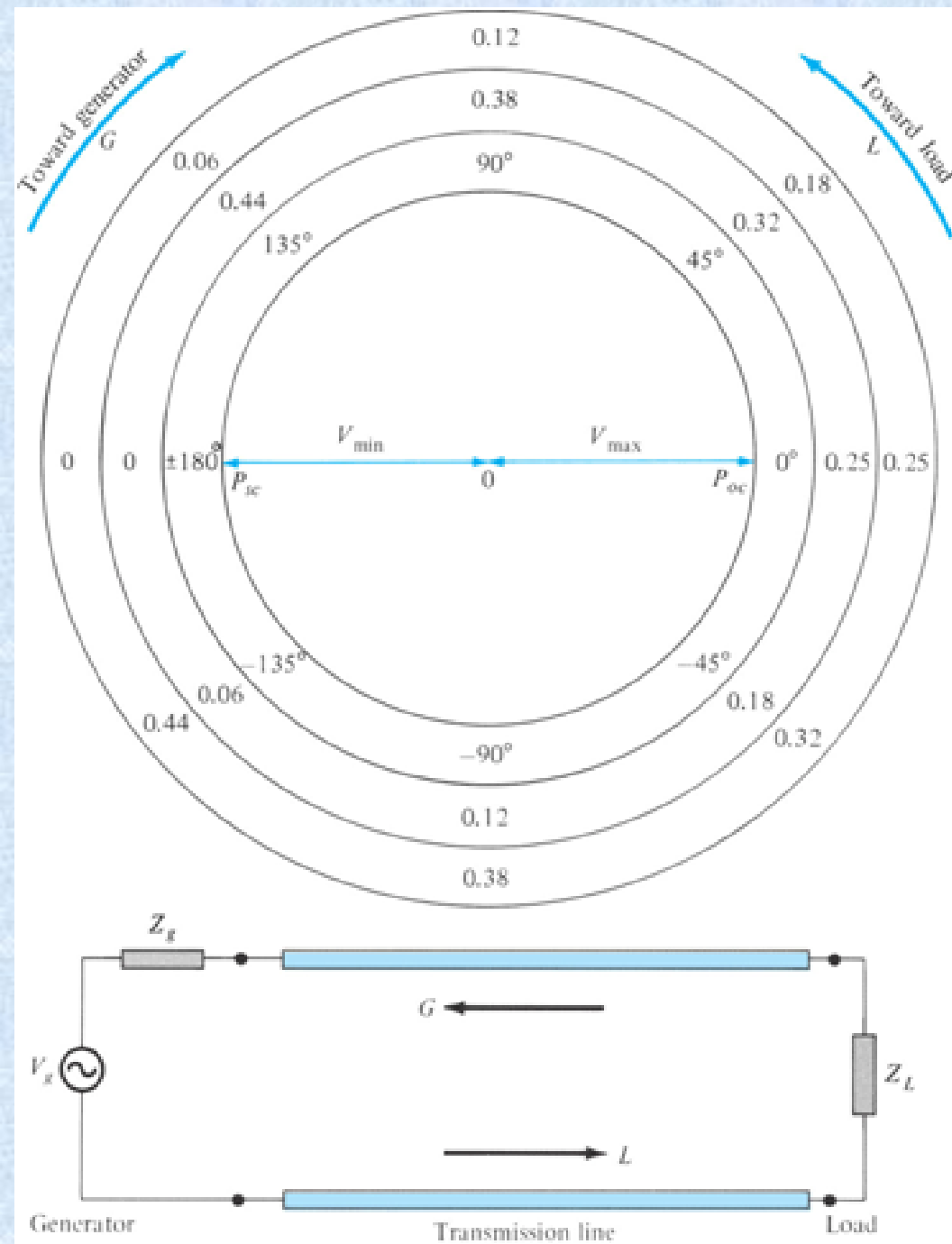


The Smith Chart

Three scales:

- **Outermost scale:** distance on line in terms of wavelength, moving toward Generator.
- **Middle scale:** distance on line in terms of wavelength, moving toward Load.
- **Innermost scale :** to determine θ_{Γ} (in degrees)

Given Z_0 , Z_L , λ and length of line, We can determine Z_{in} , Y_{in} , s , and Γ .



Example 11.4

- A lossless transmission line with $Z_0=50\ \Omega$ is 30 m long and operates at 2 MHz. The line is terminated with a load $Z_L=60+j40\ \Omega$. If $u=0.6c$ on the line, find
 - (a) The reflection coefficient Γ
 - (b) The standing wave ratio s
 - (c) The input impedance.

Solution

Method 1 (Without the Smith Chart)

$$(a) \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{60 + j40 - 50}{60 + j40 + 50} = \frac{10 + j40}{110 + j40} = 0.3523 \angle 56^\circ$$

$$(b) s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.3523}{1 - 0.3523} = 2.088$$

Example 11.4- Solution continued

$$(c) \beta l = \frac{\omega}{u} l = \frac{2\pi(2 \times 10^6)}{0.6(3 \times 10^8)} (30) = \frac{2\pi}{3} = 120^\circ \text{ (electrical length)}$$

$$\begin{aligned} Z_{in} &= Z_0 \left[\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \right] \\ &= 50 \left[\frac{60 + j40 + j50 \tan 120^\circ}{50 + j(60 + j40) \tan 120^\circ} \right] \\ &= 23.97 + j1.35 = 24.01 \angle 3.22^\circ \end{aligned}$$

Example 11.4

(a) Normalized : $z_L = \frac{Z_L}{Z_0}$

$$= \frac{60 + j40}{50} = 1.2 + j0.8$$

Locate z_L at point P

$$|\Gamma_L| = \frac{OP}{OS} = 0.3516$$

θ_r is between OS and OP :

$$\theta_r = \text{angle } POS = 56^\circ$$

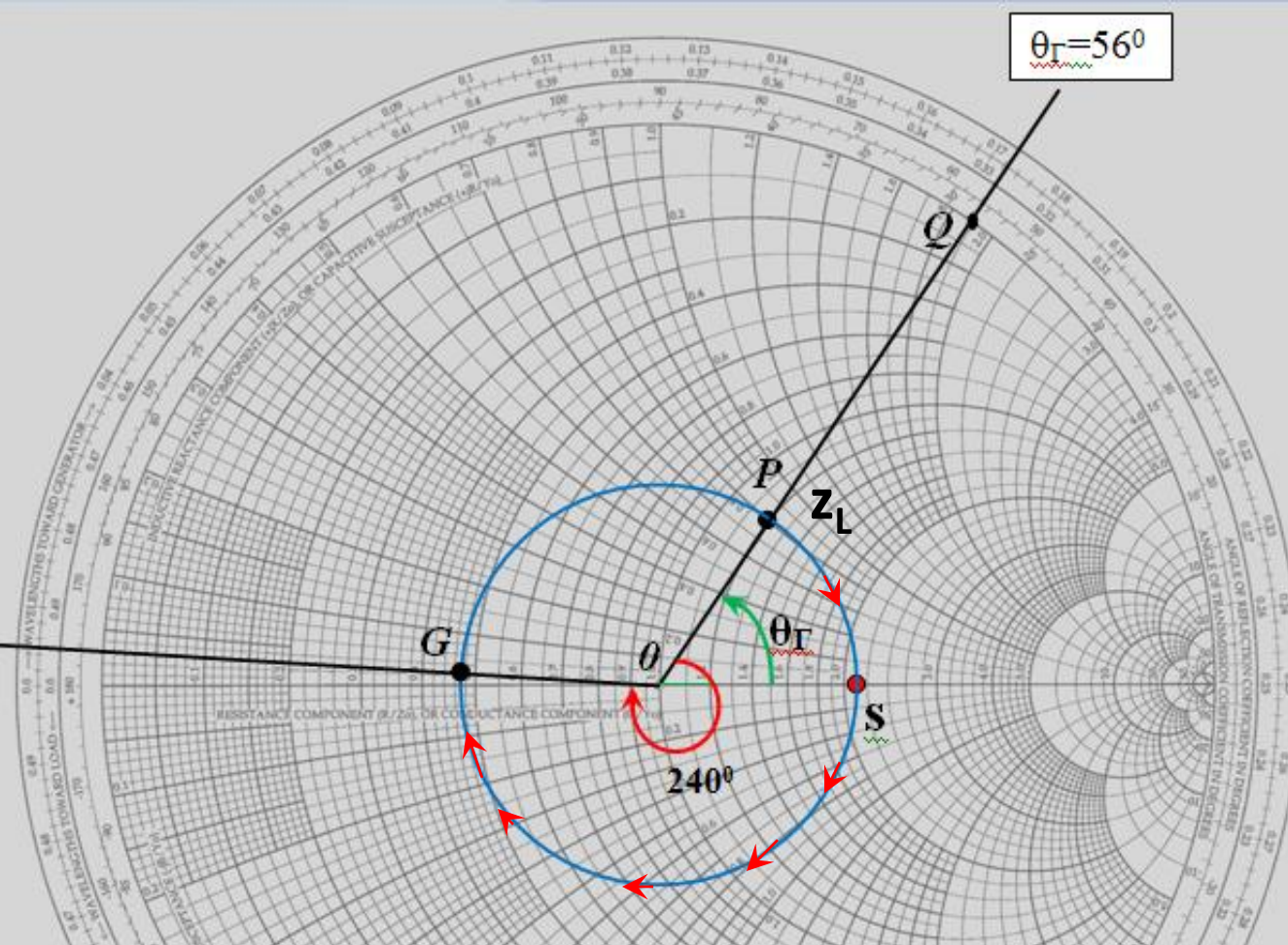
$$\rightarrow \Gamma_L = 0.3516 \angle 56^\circ$$

(c) To find Z_{in} , express l in terms of λ or in degrees.

$$\lambda = \frac{u}{f} = \frac{0.6(3 \times 10^8)}{2 \times 10^6} = 90 \text{ m}, \quad l = 30 \text{ m} = \frac{30}{90} \lambda = \frac{\lambda}{3} \rightarrow \frac{720^\circ}{3} = 240^\circ$$

$\rightarrow$ Move clockwise from load toward Generator 240° on the s-circle from point P to point G . At G , We obtain: $z_{in} = 0.47 + j0.03$

$$\text{Hence } Z_{in} = Z_0 z_{in} = 50(0.47 + j0.03) = 23.5 + j1.5 \Omega$$



(b) To find standing wave ratio s , draw s-circle (radius OP and center O)

$\rightarrow$ s-circle meets Γ_r -axis at

$$s = 2.1$$

Example 11.5

A load of $100+j150\ \Omega$ is connected to a $75\ \Omega$ lossless line. Find:

- a) The reflection coefficient Γ
- b) The standing wave ratio s
- c) The load admittance Y_L
- d) Z_{in} at 0.4λ from the load.
- e) The locations of V_{max} and V_{min} with respect to the load if the line is 0.6λ long.
- f) Z_{in} at the generator.

Example 11.5

$$(a) z_L = \frac{Z_L}{Z_0} = \frac{100 + j150}{75} \\ = 1.33 + j2 \text{ (point } P)$$

$$|\Gamma_L| = \frac{OP}{OQ} = 0.659$$

$$\theta_\Gamma = \text{angle } POS = 40^\circ$$

$$\rightarrow \Gamma_L = 0.659 \angle 40^\circ$$

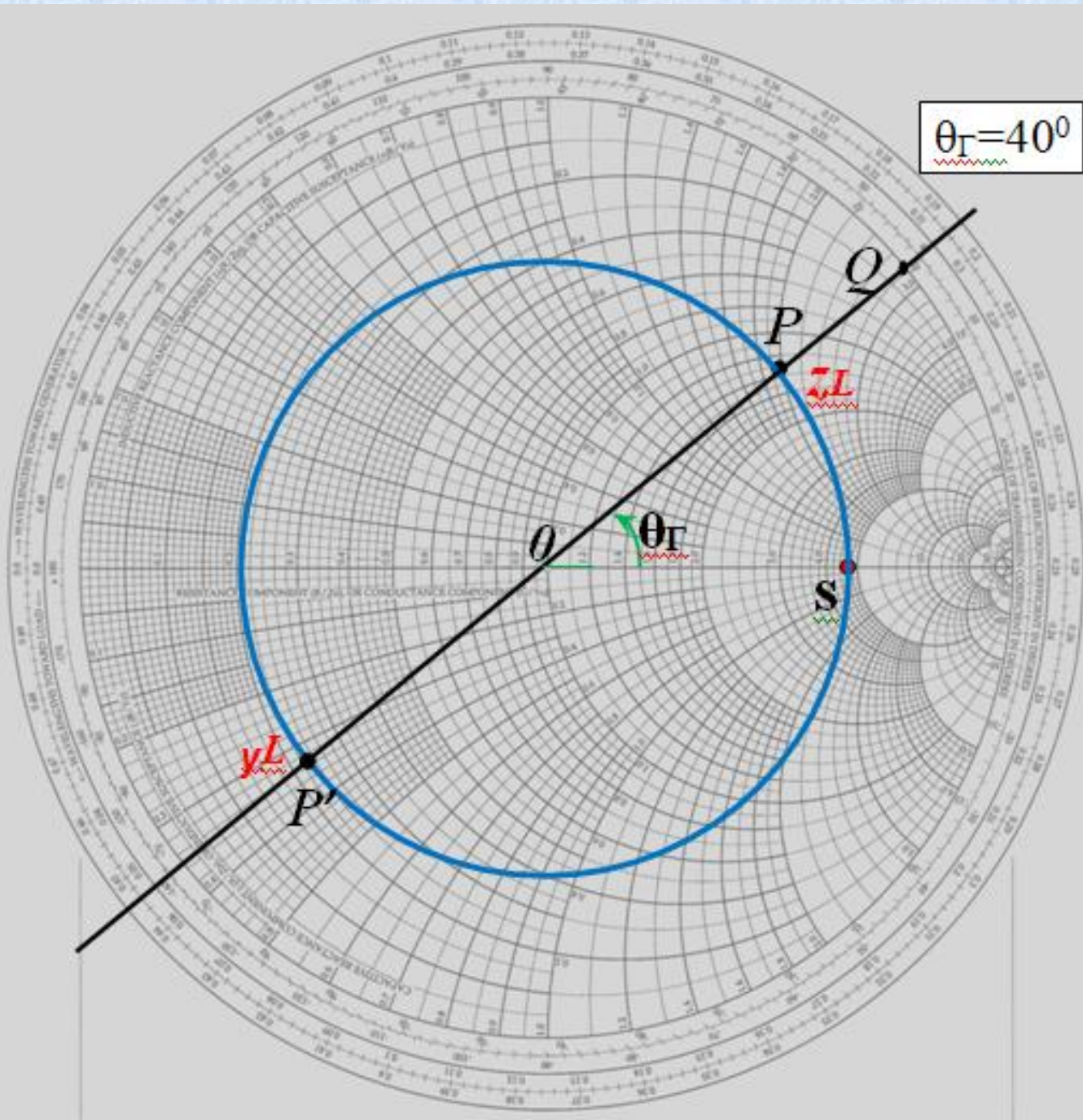
$$\text{Check: } \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_L = \frac{100 + j150 - 75}{100 + j150 + 75}$$

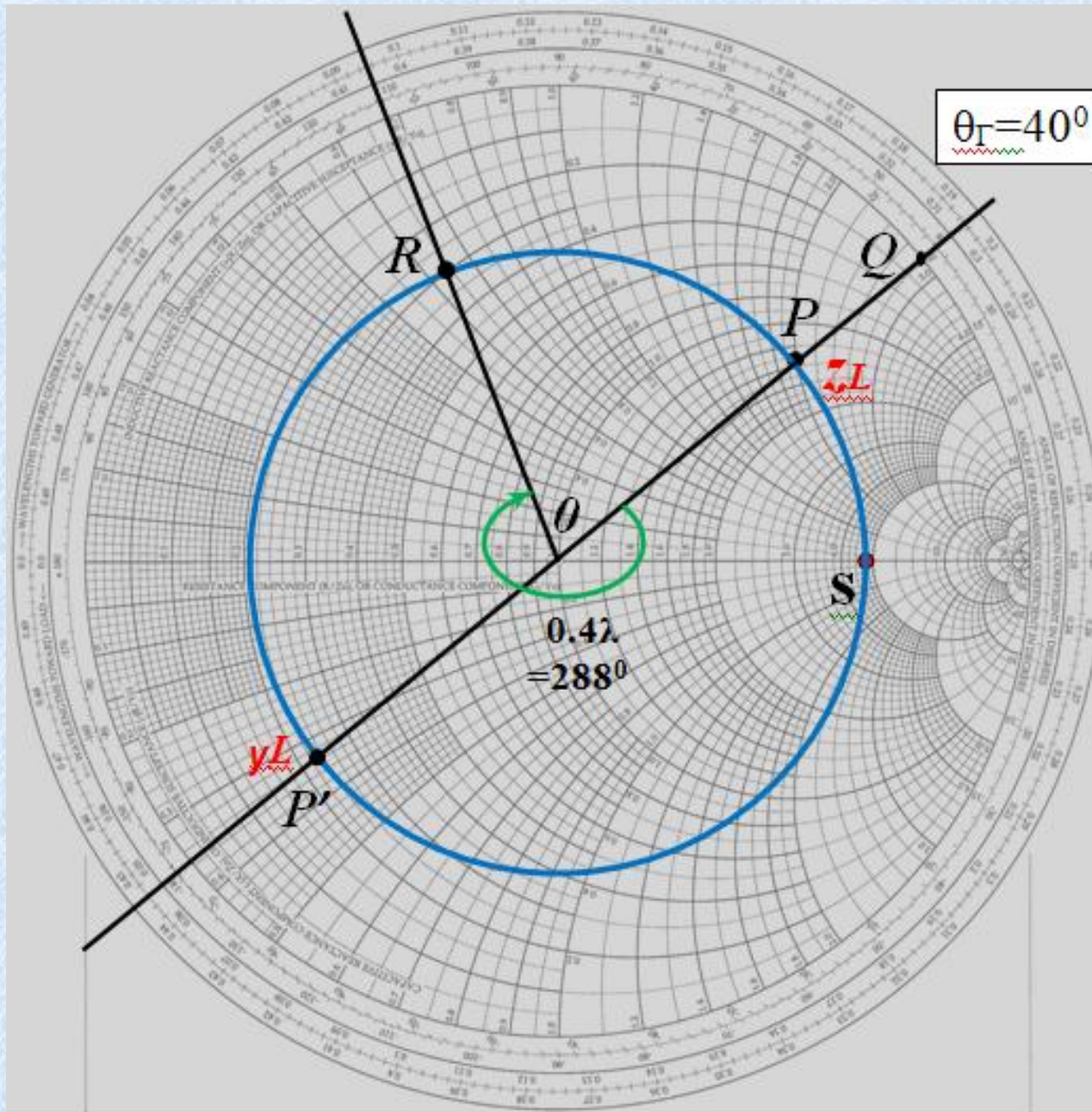
$$\Gamma_L = 0.659 \angle 40^\circ$$

(b) Draw s-circle passing through P and obtain $s=4.82$

$$\text{check: } s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = 4.865$$



Example 11.5



(c) $y_L = 0.228 - j0.35$

$$Y_L = Y_0 y_L$$

$$= \frac{1}{75} (0.228 - j0.35)$$

$$Y_L = 3.04 - j4.67 \text{ mS}$$

Check :

$$Y_L = \frac{1}{Z_L} = \frac{1}{100 + j150}$$

$$= 3.07 - j4.62 \text{ mS}$$

(d) The 0.4λ corresponds to $0.4 \times 720^\circ = 288^\circ$
 → Move 288° on s-circle from P toward generator (clockwise) to reach point R .

$$z_{in} = 0.3 + j0.63$$

$$Z_{in} = Z_0 z_{in} = 75(0.3 + j0.63)$$

$$= 22.5 + j47.25 \Omega$$

Example 11.5 – Solution continued

(d)

Check :

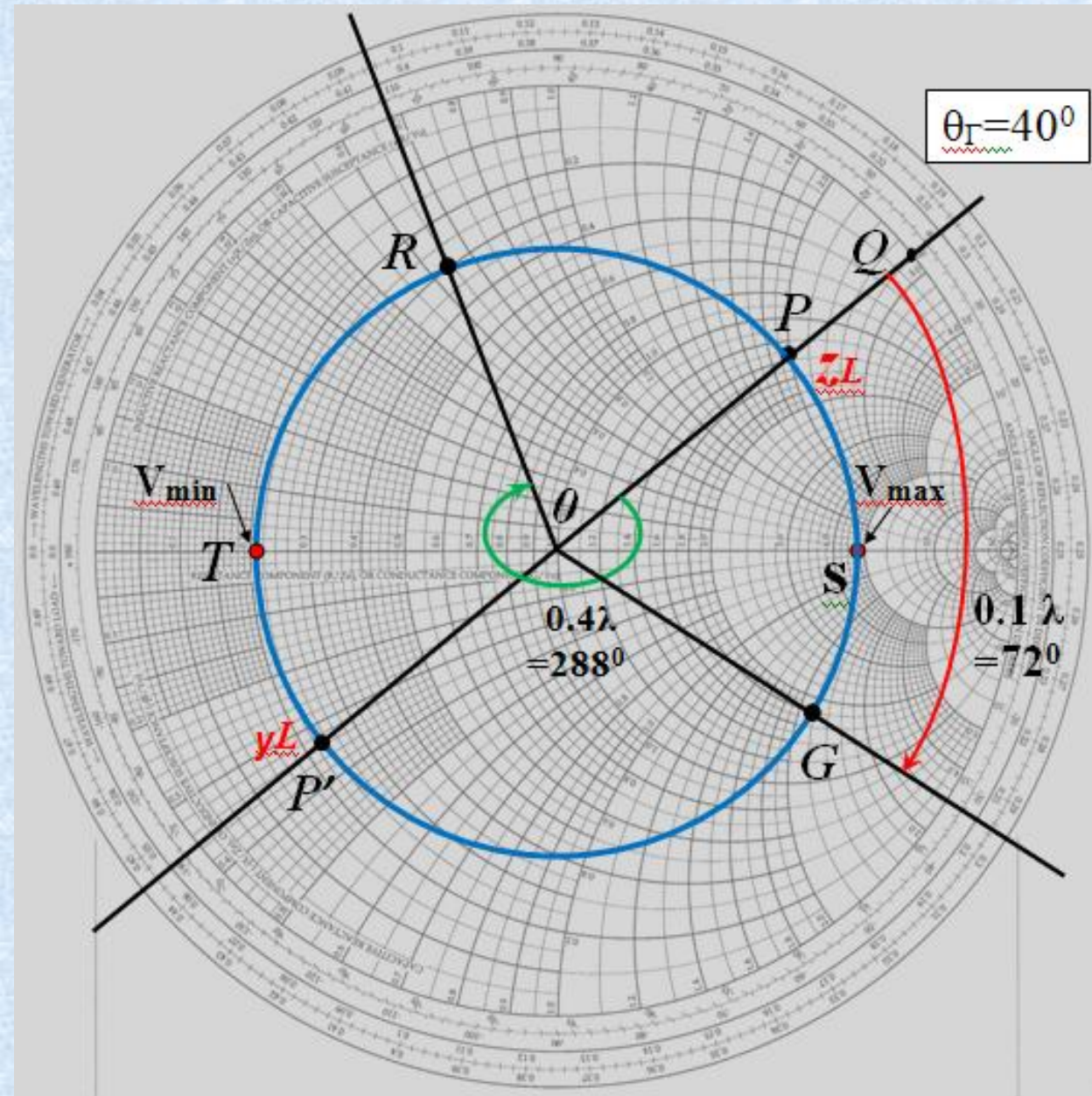
$$\beta l = \frac{2\pi}{\lambda} (0.4\lambda) = 360^\circ (0.4) = 144^\circ$$

$$\begin{aligned} Z_{in} &= Z_0 \left[\frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \right] \\ &= 75 \left[\frac{100 + j150 + j75 \tan 144^\circ}{75 + j(100 + j150) \tan 144^\circ} \right] \\ &= 54.41 \angle 65.25^\circ \end{aligned}$$

or

$$Z_{in} = 21.9 + j47.6 \, \Omega$$

Example 11.5



(e) $0.6\lambda \rightarrow 0.6 \times 720^\circ = 432^\circ$
 $= 1 \text{ revolution} + 72^\circ$

Start from P (Load), move along s-circle 432° , or one revolution $+72^\circ$, and reach Generator at point G .

→ We have passed through point T (location of $V_{\min}$) once, and point S (location of $V_{\max}$) twice. From load:

1st $V_{\max}$ is located at :

$$\frac{40^\circ}{720^\circ} \lambda = 0.055\lambda$$

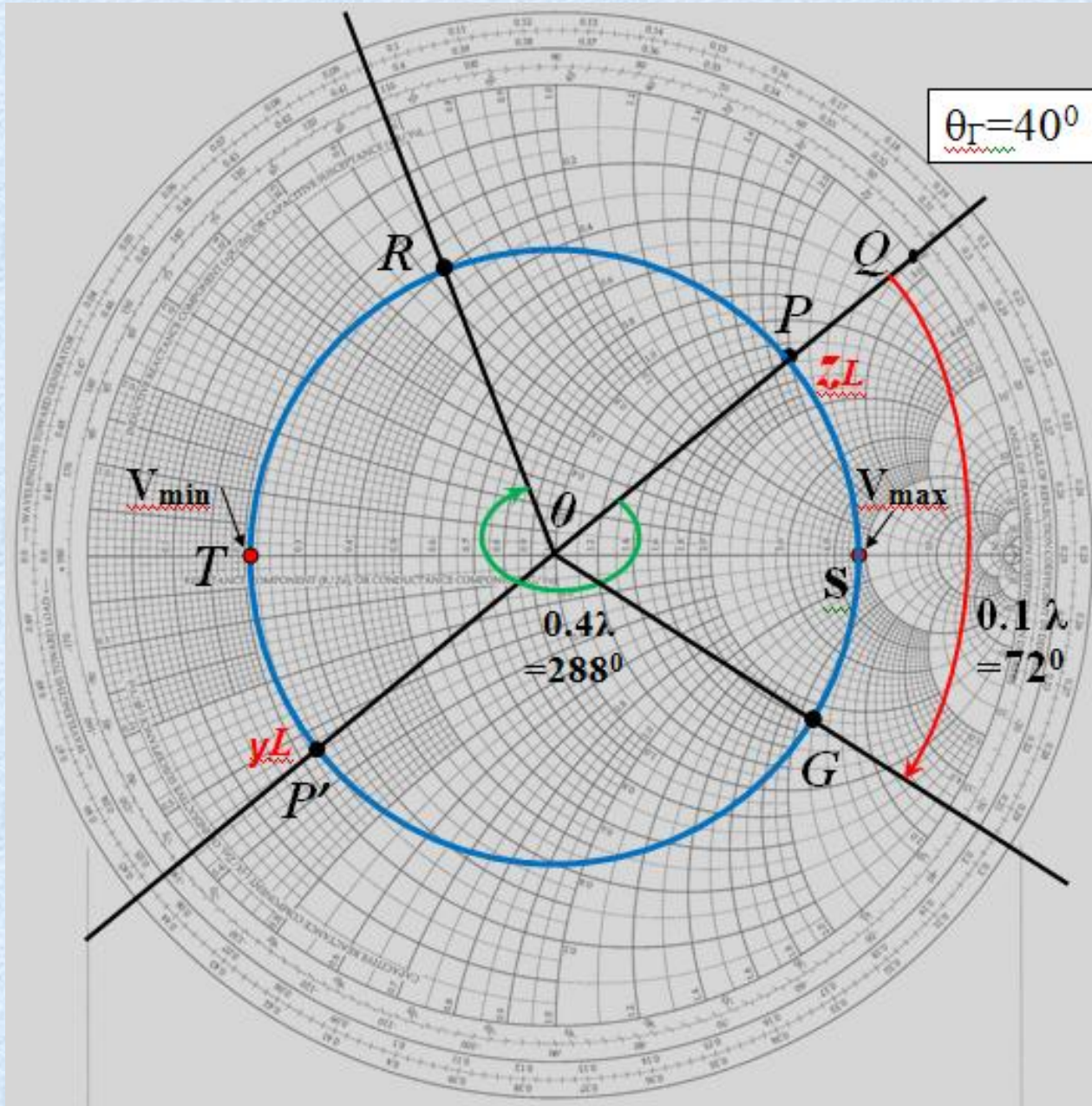
2nd $V_{\max}$ is located at :

$$0.055\lambda + \frac{\lambda}{2} = 0.555\lambda$$

The only $V_{\min}$ is located at :

$$0.055\lambda + \lambda / 4 = 0.3055\lambda$$

Example 11.5



(f) At G (Generator end),

$$z_{in} = 1.8 - j2.2$$

$$Z_{in} = 75(1.8 - j2.2) \\ = 135 - j165$$